

Matrix inequalities and norms

The Marcus–de Oliveira determinantal conjecture

Difficulty: extreme **Importance:** broadly interesting**Status:** Partially resolved

Rating rationale: This decades-old spectral enclosure conjecture for arbitrary normal pairs is a major matrix-theory barrier with consequences for determinant and spectral analysis.

Problem statement

For every integer $n \geq 1$, let $A, B \in \mathbb{C}^{n \times n}$ be normal matrices, meaning $AA^* = A^*A$ and $BB^* = B^*B$. List their eigenvalues with algebraic multiplicity as $a_1, \dots, a_n$ and $b_1, \dots, b_n$. Is

$$\det(A + B) \in \operatorname{conv} \left\{ \prod_{i=1}^n (a_i + b_{\sigma(i)}) : \sigma \in S_n \right\}?$$

Here S_n is the permutation group and the convex hull is taken in $\mathbb{C} \cong \mathbb{R}^2$.

Partial result — 13 September 2026

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[Theorem 2.1](#) gives a fourteen-term signed determinant identity for every order-four complex unitary and every pair of complex spectra. Theorems 3.1–3.2 establish optimal common negative mass on the positive-obstruction regions and show that at most one such obstruction is positive. Corollary 4.1 and Theorem 4.2 prove the original inclusion for a specified class of relative unitaries, including an operator-norm ball of radius $1/100$ around the displayed Hadamard matrix; Corollary 4.3 extends these classes by block sums. Theorem 5.1 reduces the remaining support directions, and Theorem 7.2 gives a sharp real-orthogonal obstruction bound. [Proof source](#).

The unrestricted target remains open. The manuscript does not establish the determinant inclusion for arbitrary order-four unitaries, much less all dimensions. Negative spectrum-independent weights are not counterexamples to the spectrum-dependent convex-hull assertion. The zero-obstruction classification is also left open. These are partial results under the retained original assumptions and quantifiers, not a full resolution.

The stated partial results passed a separate [independent informal Codex AI-agent audit](#), with fresh [exact and regression checks](#). No Lean verification, external human peer review or novelty claim is asserted. The original ID, path, statement and source credit are retained.

Why it matters

The conjecture would bound a determinant using only two spectra, even when the matrices are not simultaneously diagonalizable. It is a fundamental spectral enclosure problem for structured matrix computations.

References

1. A. Kovacec, *A conjecture more precise and stronger than the one by Marcus and de Oliveira*, DMUC Preprint 26-11 (7 April 2026), §1, Conjectures 1 and 1'. [Primary paper](#).
2. N. Bebiano and J. P. da Providência, *Revisiting the Marcus–de Oliveira conjecture*, Mathematics 13(5) (2025), 711, §1. [DOI](#).

3. J. I. Mulero-Martínez, *A variational framework for determinantal inequalities of normal matrices: Successes and obstructions*, *Linear Algebra and its Applications* 740 (2026), 19–38; abstract and structured-class results. [DOI](#).

Status check — 2026-09-10

The April 2026 preprint proposes a stronger conjecture rather than proving this one. The July 2026 journal paper explicitly describes the unrestricted conjecture as open and proves structured cases. Searches included *Marcus de Oliveira conjecture 2026 proof*, *Marcus Oliveira determinant counterexample*, and the exact 2026 titles. No complete proof or counterexample was located. Primary full text was checked for reference 1; reference 3's status evidence was its publisher abstract, not an independently audited proof.

Audit update (2026-09-10): Rechecked Kovacec's Conjecture 1 and the July 2026 publisher abstract of Mulero-Martínez, then searched for a full resolution. The latter explicitly retains the general conjecture while proving structured spectral classes. This is a bounded literature check, not a proof that no solution exists.